Robotics

https://en.wikipedia.org/wiki/Robotics

Robotics is an interdisciplinary branch of engineering and science that includes mechanical engineering, electronic engineering, computer science, and others.

These technologies are used to develop machines that can substitute for humans and replicate human actions. Robots can be used in many situations and for lots of purposes, but today many are used in:

- dangerous environments. Many robots are built to do jobs that are hazardous to people, such as bomb detection, defusing bombs, finding survivors in unstable ruins, and exploring mines and shipwrecks.

- manufacturing processes

- where humans cannot survive (e.g. in space, under water, in high heat, and clean up and containment of radiation)

- in STEM (science, technology, engineering, and mathematics) as a teaching aid.

- in medicine and human health. The advent of nanorobots, microscopic robots that can be injected into the human body.

Robots can take on any form but some are made to resemble humans in appearance. This is said to help in the acceptance of a robot in certain replicative behaviors usually performed by people.

https://en.wikipedia.org/wiki/Robotic_sensing

Robotic sensing is a subarea of robotics science intended to give robots sensing capabilities, so that robots are more human-like. Robotic sensing mainly gives robots the ability to see, touch, hear and move and uses algorithms that require environmental feedback.

https://www.futurelearn.com/courses/robotic-future/0/steps/29368

The most popular language in robotics is probably C/C++ (C++ is an object-oriented successor to the C language). Python is also very popular due to its use in machine learning and also because it can be used to develop ROS packages.

- C programming and the Arduino microcontroller
- Python and the Raspberry Pi

Robot Operating System (ROS) is a set of software libraries and tools that helps you build robot applications. You can also write your own programs for ROS e.g. in C/C++ or Python.